

Foundational Assumptions in Causal Inference

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Summary

We all have heard “Correlation does not imply causation,” but how do we tell when causation is present? We face challenges such as describing causal variables and controlling confounders, and need a framework for describing and working with them.

Causal effect can be studied with the help of three foundational assumptions: exchangeability, consistency, and positivity. All of these assumptions rely greatly on knowledge about the data and its setting.

Using binary outcome, intervention, and confounder variables, this report will describe the importance of these assumptions and how to use them to describe the causal effect of the intervention on the outcome.

Introduction

If we take a large group of people who received a heart transplant, and see that 30% of them died in a certain time frame after the transplant occurred, and another large group of people who did not receive a heart transplant and had a 70% proportion of death in the same period, we would be compelled to state that the transplant caused a difference in how many people died.

However, the composition of each group matters when determining the cause of that difference. What if the transplant wasn't a cause of reduced death but it was correlated with something that reduced death? How can we narrow in on the causal effect of heart transplants on death? This is the primary task of causal inference.

Notation

Consider a setting with a binary outcome, intervention, and confounder. We will use $Y = 0$ and $Y = 1$ to denote outcomes, $A = 0$ and $A = 1$ to denote interventions, and $L = 0$ and $L = 1$ to denote prognostic factors used in assignment of A . L is also a confounder of A and Y .

In our heart transplant example, we will define $A = 0$ as no transplant, $A = 1$ as transplant, $Y = 0$ as being alive at a set amount of time after intervention, and $Y = 1$ as dying within that set period after intervention. $L = 0$ will refer to not being in critical condition at time of treatment and $L = 1$ will refer to being in critical condition at time of treatment. One can infer that an individual's level of L will affect their probability of dying, and that somebody making treatment decisions for patients would be more inclined to assign somebody with $L = 1$ a heart transplant than somebody with $L = 0$.

In a real study, the information we have is each individual's levels of Y and A , or $Y|A$, that is, their value of Y that has happened while they received treatment A . We would like to know what would happen to an individual in each level of A . If somebody doesn't get a transplant, we would like to know what would have happened if they did receive a transplant. It's not possible to know their outcome under all levels of A , because only one can happen. This is known as the foundational problem of causal inference.

To begin exploring the unknown scenario, we need a term for what would happen in the case that the other treatment level was chosen.

Definition: Counterfactual outcome

The outcome that an individual would have in the case that they have a certain intervention, regardless of if they actually had that intervention. The outcome Y if an individual experienced intervention a is Y^a .

$P[Y^a = 1]$ is the proportion of individuals who will experience $Y = 1$ when intervened to have $A = a$. $E[Y^a]$ is the expected value of Y under an intervention where $A = a$. When Y is binary, they are the same, as $P[Y^a = 1] = 1 * P[Y^a = 1] + 0 * P[Y^a = 0] = \sum_{y=0}^1 yP[Y^a = y] = E[Y^a]$.

The notation can be difficult to parse, because there are many pieces of information included. To give an example of how it can be read, $P[Y^1 = 0]$ refers to the probability that outcome Y will equal 0 in the event that intervention A has a value of 1.

Quantities of Interest

Causal effect can be measured in many ways. There are many more than we will discuss, but here are three causal effect measures detailing the effect of changing $A = 0$ to $A = 1$:

Causal Risk Difference: $P[Y^{a=1} = 1] - P[Y^{a=0} = 1]$

Causal Risk Ratio: $\frac{P[Y^{a=1}=1]}{P[Y^{a=0}=1]}$

Causal Odds Ratio: $\frac{P[Y^{a=1}=1]/P[Y^{a=1}=0]}{P[Y^{a=0}=1]/P[Y^{a=0}=0]}$

Causal risk difference is a difference in probabilities. If the causal risk difference is 0.1, then when $a = 1$, $P[Y = 1]$ is 0.1 greater than when $a = 0$.

The causal risk ratio is a ratio between probabilities. If the causal risk ratio is 0.1, then $P[Y^{a=1} = 1]$ is one-tenth of $P[Y^{a=0} = 1]$, so treatment $A = 1$ cuts risk down to a tenth of what it was.

Causal risk difference and causal risk ratio describe different things. This is illuminated by an example. Imagine a treatment that brings risk, or $P[Y = 1]$, from 90% down to 80%. The causal risk difference is 10%, and the causal risk ratio is $\frac{80}{90} = 0.89$, so risk under treatment is 89% that of under no treatment.

Imagine another case where a treatment brings risk from 11% to 1%. The causal risk difference is still 10%, but the causal risk ratio is $\frac{1}{11} = 0.09$, so risk under treatment is 9% that of under no treatment. In these two cases, the risk difference is the same, but the risk ratio is much different.

Odds refers to the probability of something happening ($P[Y = 1]$) divided by the probability of it not happening ($P[Y = 0]$). The ratio in question is the proportion between odds when $a = 1$ and odds when $a = 0$.

To justify our claims about causal effect, we need to believe that our population and their interventions satisfy important assumptions. Here are three of those assumptions.

Exchangeability

Definition: Exchangeability $Y^a \perp\!\!\!\perp A$

If intervention A is exchangeable, A is independent of Y^a . Each individual could have their values of A swapped and the observed effect of A would remain the same.

Exchangeability $Y^a \perp\!\!\!\perp A$ refers to situations where swapping the assigned treatment of each individual makes no difference in the observed causal effect. This is a required assumption for observing causal effect, and, in an experimental study, is most commonly satisfied by treatment being assigned randomly.

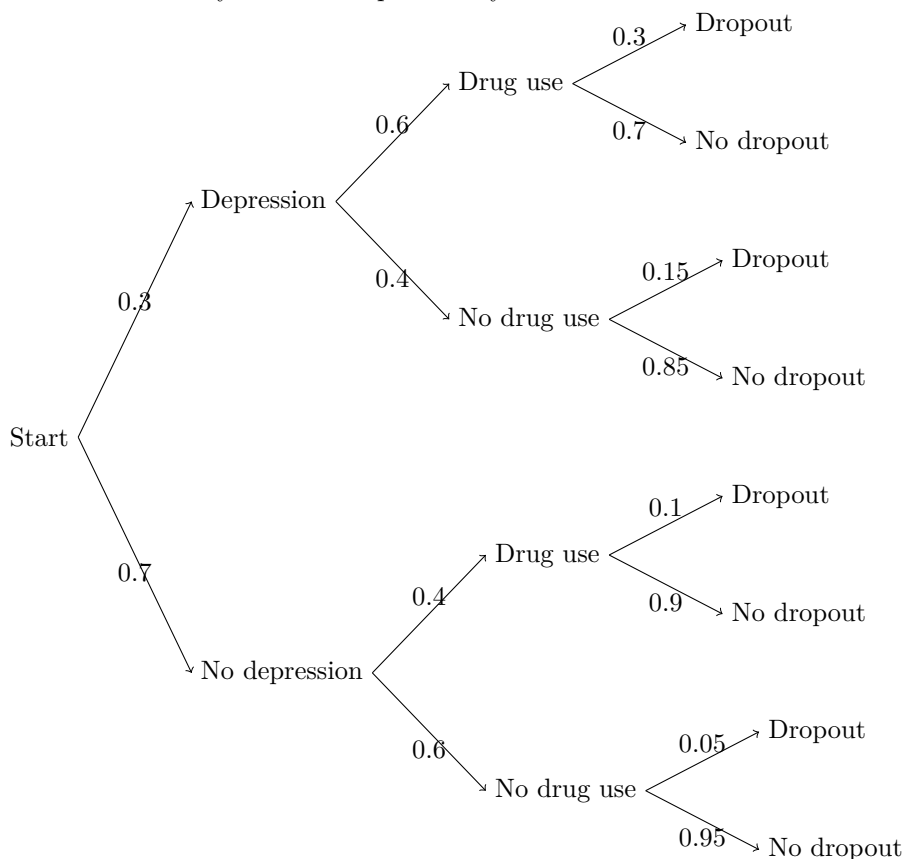
Exchangeability means $P[Y^a = 1|A = 1] = P[Y^a = 1|A = 0]$ for any a . That is, the counterfactual outcome for treatment a in the group of people who had treatment $A = 1$ has the same expectation as in the group who had treatment $A = 0$.

In our heart transplant example, the transplants were assigned randomly with each patient having a 30% chance of receiving a transplant. Because the treatments are random, we can assume that the distribution of patients' counterfactual outcomes is similar between the $A = 0$ and $A = 1$ groups. In a large enough sample, the observed expected outcome for each treatment would be roughly equal to what it would be if the groups were switched, and the 30% of patients who received the transplant didn't receive one and the other 70% did receive a transplant. So, exchangeability holds.

Now, let's consider a population of 15 year olds and outcome dropping out of high school. For A , we will consider current use of drugs, where $A = 0$ means no drug use and $A = 1$ means drug use.

This scenario is complex for many reasons. The first complexity we'll study is the idea that another variable could affect both use of drugs and dropout, swaying the count of dropouts who used drugs. Let's consider depression as this confounder, with $L = 1$ describing a child who scores above a certain threshold on a depression questionnaire and $L = 0$ describing a child who does not.

Here is an arbitrary conditional probability tree for these variables.



This conditional probability tree can be read by treating the number on each branch as the probability of the variable on the right given all of the variables on the left. The topmost branches mean this: $P(\text{Depression}) = 0.3$, $P(\text{Drug use}|\text{Depression}) = 0.6$, and $P(\text{Dropout}|\text{Depression, Drug use}) = 0.3$. We can

multiply these numbers to find $P(\text{Depression, Drug Use, Dropout}) = 0.3 * 0.6 * 0.3 = 0.054$.

Let's return to A, Y , and L . For exchangeability to hold, we need $P(Y^{a=1} = 1|A = 1) = P(Y^{a=1} = 1|A = 0)$. From this tree, we can show that $P(Y^{a=1} = 1|A = 1) = 0.178$ and $P(Y^{a=1} = 1|A = 0) = 0.144$. This means that the true value of A is associated with the counterfactual outcomes $Y^{a=1}$, so we cannot find the overall risk $P(Y^{a=1} = 1)$ and A is not exchangeable.

Within the levels of L , assignment of A is still not random, because there are many further confounders for drug use and dropout.

If we return to the heart transplant example, we have a situation where we can use a confounder variable to find conditional exchangeability.

Take $L = 1$ to be critical condition and $L = 0$ to be non-critical condition. Even in an experimental study, because of the limited number of hearts, doctors want patients with $L = 1$ to have a higher chance of getting a heart transplant to improve overall death rates. They assign $P[A = 1|L = 1] = 0.8$ and $P[A = 1|L = 0] = 0.4$. Now, L affects both assignment of A and outcome Y . Because A is random within levels of L , A is exchangeable given the level of L .

Because this conditional exchangeability holds, $P(Y^{a=1} = 1|A = 1, L = l) = P(Y^{a=1} = 1|A = 0, L = l) = P(Y^{a=1}|L = l)$ and we can find the risk of death for each level of L .

Definition: Conditional Exchangeability $Y^a \perp\!\!\!\perp A|L$

If intervention A is conditionally exchangeable, A is independent of Y^a given that $L = l$. Each individual could have their values of A swapped and the observed effect of A within each level of L would remain the same.

Consistency

Definition: Consistency

The outcomes observed are linked to a well-defined treatment. For every outcome Y , if A is set to value a , $Y = Y^a$.

Consistency links the counterfactuals to the observed data, and lets us assume that $Y|A$ is the same as $Y^a|A$ which, with exchangeability, is the same as Y^a . Without consistency, we would not be able to make conclusions about the causal effect of an intervention because we would not be able to sort our data into levels of that intervention.

A well-defined intervention is one that corresponds to actions that can be described in some level of detail and carried out accordingly. There should be minimal doubt about what counts as each level of the

intervention.

It's when the researchers agree that the groups of events that could take place after an individual is assigned intervention $A = a$ can be grouped together into the label $A = a$ and still lead to a meaningful measure of causal effect.

Here is an example to explain why defining interventions requires decisions made by researchers. A well-defined heart transplant can look different ways. Take one study in which detailed instructions are given for each step of the surgery: pre, post, and during the procedure. Take another study in which the intervention is defined as this: the patient gets a heart transplant at one of the hospitals the study covers.

Both have merit. Giving detailed instructions for the transplant controls, to a degree, what exactly the intervention is, and estimates the causal effect of the ideal form of that intervention.

Outside of a randomized experiment, a heart transplant doesn't have specifications past the hospital's policies, the surgeon's training, and other random factors. If a study is on the general intervention heart transplant versus no heart transplant and not a specific practice for performing heart transplants, the latter, unguided intervention would be a realistic portrayal of the effect of assigning a patient a heart transplant. The latter approach also allows for a study to be observational rather than experimental, unlike the former approach.

Since the two experiments are measuring the effect of slightly different treatments, the observed causal effect will be different.

There isn't any level of instruction that can make the transplants identical in the way the intervention was carried out, so the two examples are at different places on the same scale.

In the adolescent drug use example, there's a question of what defines 'drug use'. Would a teen using any amount or type of drug in any time frame be described with $A = 1$ or should the category be more restricted?

A study that doesn't specify the intervention and ensure that the data uses that specification will get confused results. Data collection might classify kids who used drugs once in the past year as either $A = 0$ or $A = 1$, with no rule to separate the levels. Without a clearer definition of drug use, similar inconsistencies would happen with any other grey area, such as kids who used drugs in the past but not currently. This is a consistency violation, because outcomes for some individuals will not clearly be under one of $A = 0$ and $A = 1$.

Positivity

Definition: Positivity

For all values of l with $P[L = l] > 0$ in the population of interest, $P[A = a|L = l] > 0$ for all a .

Positivity means that for each treatment and for each subset of the population, determined by levels of L , there must be at least one individual whose probability of the treatment is more than 0. This assumption allows causal effect to be measured for each group.

In a case where positivity doesn't hold, it's easy to show that causal effect can't be calculated.

Take Y as death, A as transplant, and L as critical condition. If transplants were only given to those in critical condition, $E[Y^1|L = 0]$ (average outcome under transplant for those not in critical condition) could not be calculated, so none of the effect measures could be calculated.

This violation may be structural or practical in nature. If the violation happens because of a structural fact about the population and the application of the intervention, then no matter how large the population, nobody with $L = l$ for some l will have a certain treatment a .

There could be a hospital policy that only patients in critical condition can get heart transplants. This means that in the study, no matter how large the sample of individuals, none in the non-critical group will get a heart transplant, effectively cutting them out of the study.

Another case is that of random chance. If a sample is small, it's possible that no individual in a group with $L = l$ will be randomly assigned treatment $A = a$. This is a limit set by the impracticality of running an experiment with heart transplants. They are a relatively uncommon medical procedure, so if a study is observational, there might not be an example of an individual not in critical condition who received a heart transplant.

Conclusion

Exchangeability, consistency, and positivity provide the foundation for identifying causal effects from observed data.

Many treatments cannot be studied in a randomized trial, because they would be unethical, impractical, or impossible to intervene on. For example, it would be unethical to have teenagers do drugs for the sake of a study. An observational study, where interventions are not assigned by the investigators, is a logical next choice.

As shown in the adolescent drug use example, observational studies have added challenges. Exchangeability isn't easily confirmed. We separated the population by Depression, letting us adjust for differences

in dropout likelihood in those who scored differently on a depression questionnaire. However, there may be many more confounding factors that affect values of A .

This doesn't mean that observational studies can never assume exchangeability because of endless unmeasurable confounders. Investigators need to agree that the condition is reasonably justified and acknowledge that it may lead to imperfect results.

Similar problems happen with consistency. The subjects of observational studies are not interfered on, or are even far in the past compared to the people investigating them, so it's difficult to sort what happened with each individual into well-defined intervention.

Positivity can also pose problems. If the data available does not have examples of individuals from a certain subset having some treatment, there is not much to be done about finding outcome data for individuals in that subset under that treatment.